Scalability & Future Plan

Date	02 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document describes the scalability strategy and future enhancement roadmap for the OptiCrop system. The goal is to ensure that the platform can support a larger number of users, process increasing amounts of agricultural data, and incorporate advanced features that improve farming productivity and decision-making.

Current System Limitations:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users on local deployment	Deploy on cloud infrastructure with load balancing support
2	Data Storage	Uses local datasets and model files	Integrate cloud databases and centralized data storage
3	Performance	Single-server Flask application	Optimize model serving and implement caching mechanisms
4	Security	Basic input validation and request handling	Add authentication, authorization, and secure API access

Scalability Plan:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)

1	Limited to crop recommendation only	Farmers cannot receive fertilizer or yield recommendations	High
2	No real-time weather integration	Predictions do not consider current weather conditions	High
3	Web application only	Accessibility is limited for users preferring mobile applications	Medium

Future Roadmap:

Phase	**Planned Feature / Enhancement**	**Target Timeline**	**Expected Impact**
Phase 2	Weather Forecast Integration	Next 3 Months	Improves recommendation accuracy using real-time climate data
Phase 3	Fertilizer Recommendation System	Next 6 Months	Provides farmers with nutrient management guidance
Phase 4	Mobile Application and IoT Integration	Next 6 Months	Enables remote monitoring, wider accessibility, and smart farming support